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Bregman distances without coercive condition: suns, Chebyshev sets and Klee sets

Xian-Fa Luo^a, Li Meng^b, Ching-Feng Wen^c and Jen-Chih Yao^d

^aDepartment of Mathematics, China Jiliang University, Hangzhou, People's Republic of China;

^bDepartment of Mathematics, Zhejiang University of Technology, Hangzhou, People's Republic of China; ^cCenter for Fundamental Science, Kaohsiung Medical University, Kaohsiung, Taiwan; ^dCenter for General Education, China Medical University, Taichung, Taiwan

ABSTRACT

In a real Banach space X , we introduce for a non-empty set C in X the notion of suns in the sense of Bregman distances and show that C is such a sun if and only if C is convex. Also, we give some necessary and sufficient conditions for a compact set to be the Klee set, extending corresponding results on the Euclidean space.

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1. Introduction

Let X be a real Banach space, and let $f : X \rightarrow \overline{\mathbb{R}} := \mathbb{R} \cup \{+\infty\}$ be a proper convex function with its domain $\text{dom} f$. Then the Bregman distance with respect to f between the points $x, y \in \text{dom} f$ is defined by

$$D_f(x, y) := f(x) - f(y) - f'_+(y, x - y), \quad (1)$$

where $f'_+(y, v)$ is the directional derivative of f at y in the direction $v \in X$, i.e.

$$f'_+(y, v) := \lim_{t \rightarrow 0^+} \frac{f(y + tv) - f(y)}{t}.$$

In particular, when f is Gâteaux differentiable on $U := \text{int}(\text{dom} f) \neq \emptyset$ with its Gâteaux derivative denoted by ∇f , then (1) is derived to

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle \quad \text{for each } (x, y) \in \text{dom} f \times U.$$

For a non-empty closed set $C \subseteq U$, following [1–4], define the left Bregman distance function and, assuming C is bounded, the left Bregman farthest distance

function of C by

$$\overleftarrow{D}_C^f(y) := \inf_{x \in C} D_f(x, y) \quad \text{and} \quad \overleftarrow{F}_C^f(y) := \sup_{x \in C} D_f(x, y) \quad \text{for each } y \in U, \quad (2)$$

respectively. Thus, the left Bregman projection and the left Bregman farthest point map onto C are defined accordingly by

$$\overleftarrow{\Pi}_C^f(y) := \left\{ x \in C : D_f(x, y) = \overleftarrow{D}_C^f(y) \right\} \quad (3)$$

and

$$\overleftarrow{Q}_C^f(y) := \left\{ x \in C : D_f(x, y) = \overleftarrow{F}_C^f(y) \right\} \quad \text{for each } y \in U. \quad (4)$$

Recall from [1,2] (resp., [3,4]) that C is left Chebyshev (resp., left Klee) with respect to the left Bregman distance, or simply $\overleftarrow{D}_f$ -Chebyshev (resp., $\overleftarrow{D}_f$ -Klee), if $\overleftarrow{\Pi}_C^f(y)$ (resp., $\overleftarrow{Q}_C^f(y)$) is a singleton for each $y \in U$. Motivated by the classical convexity problem of a Chebyshev set and the single-point problem of a Klee set in Hilbert spaces, Bauschke et al. first proposed and studied the same problems in $\mathbb{R}^n$ but in the sense of Bregman distances, i.e. the convexity problem of a left Chebyshev set and the single-point problem of a left Klee set with respect to Bregman distances in $\mathbb{R}^n$ in [1,3], respectively. They shown in [1] that if f is a convex function of Legendre type and 1-coercive, then each non-empty closed subset C of U is left Chebyshev if and only if C is convex, and shown (with the same assumption for f as in [1]) in [3] that each left Klee set C in U is a singleton. Hereafter, Li et al. and Fang et al. extended and improved those results as well as many other results in [1,3] to the case of real reflexive Banach spaces, respectively (see [2,4]).

Since $D_f(\cdot, \cdot)$ is not symmetric in general, we need to define the right Bregman distance function and, assuming C is bounded, the right Bregman farthest distance function of C by

$$\overrightarrow{D}_C^f(x) := \inf_{y \in C} D_f(x, y) \quad \text{and} \quad \overrightarrow{F}_C^f(x) := \sup_{y \in C} D_f(x, y) \quad \text{for each } x \in U, \quad (5)$$

respectively. Thus similar problems for the right Chebyshev set and the right Klee set are, respectively, also explored in $\mathbb{R}^n$ in [1,3] by Bauschke et al. under the assumption that f is a convex function of Legendre type on $\mathbb{R}^n$ with full domain, and some sufficient conditions for $\nabla f(C)$ (resp., the right Klee set C) to be convex (resp., a singleton) were given. We see that the 1-coercivity assumption for f (resp., the dual function f^* of f) plays a fundamental role in the study mentioned above.

In particular, the convexity problem of a Chebyshev set and the single-point problem of a Klee set considered here are reduced to the classical ones, which have been studied extensively; see, for example [5–8] and references therein.

In this paper, we will continue to investigate the proceeding two problems in the sense of Bregman distances without the 1-coercivity assumption in general real Banach spaces. The organization of the present paper is as follows. In Section 2, we review some basic notions concerning convex analysis and give relations among these notions. In Section 3, for the left (resp., right) Bregan distance functions of C , we give the relationships among the convexity, the $\overleftarrow{D}_f$ -Chebyshev (resp., $\overrightarrow{D}_f$ -Chebyshev) property, and the $\overleftarrow{D}_f$ -solarity (resp., $\overrightarrow{D}_f$ -solarity) (see Definition 3.5) for the set C (see Theorems 3.7, 3.9 and 3.12); in particular, for a bounded compact set $C \subseteq U$, we show that C is convex if and only if C is $\overleftarrow{D}_f$ -Chebyshev provided that f has nice property enough, which extends [1, Theorem 6.6(i) $\iff$ (ii)] to the setting of the present paper. In Section 4, we give some sufficient and necessary conditions for a left (resp., right) Klee set (see, Definition 4.1) to be a singleton; see Theorem 4.6 (resp., Theorem 4.7), which is an extension of [3, Corollary 4.5] and [4, Theorem 4.2(i) $\iff$ (ii)] (resp., [3, Theorem 7.2]). As applications, the corresponding results in [4,5,6] are obtained in Section 5 as direct consequences.

2. Preliminaries

Let X be a real Banach space with its dual X^* . For $x \in X$ and $\delta > 0$, we denote by $\mathbf{B}(x, \delta)$ the closed ball centred at x with radius δ . Let C be a non-empty subset of X . Then the closure (resp., convex hull, set of extreme points) of C is denoted by $\overline{C}$ (resp., $\text{co } C$, $\text{ext } C$). For $x \in X$ and $x^* \in X^*$, by $\langle x^*, x \rangle$ we denote the value of the functional x^* at x , i.e. $\langle x^*, x \rangle = x^*(x)$. In what follows, we always assume that $f: X \rightarrow \overline{\mathbb{R}}$ be a proper lower semicontinuous convex function with its domain denoted by $\text{dom } f$. Let $x \in \text{dom } f$. The subdifferential of f at x is defined by

$$\partial f(x) := \{x^* \in X^* : f(x) + \langle x^*, y - x \rangle \leq f(y) \text{ for each } y \in X\};$$

while the conjugate function of f is the function $f^*: X^* \rightarrow \overline{\mathbb{R}}$ defined by

$$f^*(x^*) := \sup\{\langle x^*, x \rangle - f(x) : x \in X\}.$$

By [9, Theorem 2.3.3], f^* is the proper weak* lower (hence lower) semicontinuous convex function on X^* . Moreover, by [9, Theorem 2.4.2(iii)], the Young–Fenchel inequality holds:

$$\langle x^*, x \rangle \leq f(x) + f^*(x^*) \quad \text{for each pair } (x^*, x) \in X^* \times X, \quad (6)$$

and the equality holds if and only if $x^* \in \partial f(x)$, i.e.

$$\langle x^*, x \rangle = f(x) + f^*(x^*) \iff x^* \in \partial f(x) \quad \text{for each pair } (x^*, x) \in X^* \times X. \quad (7)$$

Furthermore, define $f^{**} = (f^*)^*$. Then

$$f^{**}(J_X(x)) = f(x) \quad \text{for each } x \in X \quad (8)$$

by [10, Fact 2.13], where and in the sequel, J_X is the canonical embedding map from X to X^{**} . The domain of ∂f is denoted by $\text{dom}(\partial f)$ and defined by

$$\text{dom}(\partial f) := \{x \in \text{dom } f : \partial f(x) \neq \emptyset\}.$$

Recall that the Bregman distance with respect to f is defined by (1)

Remark 2.1: There is another manner (cf. [11]) to define the Bregman distance with respect to f :

$$D_f(x, y) := f(x) - f(y) + f'_+(y, y - x) \quad \text{for any } x, y \in \text{dom } f.$$

Clearly, they coincide for any y where f is Gâteaux differentiable.

According to [12], we define the modulus of total convexity at y by

$$\nu_f(y, t) := \inf\{D_f(x, y) : x \in \text{dom } f, \|x - y\| = t\} \quad \text{for each } t \geq 0.$$

For our study, we need to introduce the locally uniform modulus of total convexity at y , which is defined by

$$\nu_f^{loc}(y, t) := \lim_{\delta \rightarrow 0^+} \inf\{\nu_f(u, t) : u \in \mathbf{B}(y, \delta) \cap \text{dom } f\} \quad \text{for each } t \geq 0.$$

Definition 2.1: Let $y \in \text{dom } f$. The function $f : X \rightarrow \overline{\mathbb{R}}$ is said to be

- (a) totally convex at y if its modulus is positive on $(0, \infty)$, i.e. $\nu_f(y, t) > 0$ for each $t > 0$;
- (b) locally uniformly totally convex at y if its locally uniform modulus is positive on $(0, \infty)$, i.e. $\nu_f^{loc}(y, t) > 0$ for each $t > 0$;
- (c) essentially strictly convex if $(\partial f)^{-1}$ is locally bounded on its domain and f is strictly convex on every convex subset of $\text{dom}(\partial f)$;
- (d) essentially smooth if ∂f is both locally bounded and single-valued on its domain;
- (e) Legendre if f is both essentially strictly convex and essentially smooth.

Remark 2.2: (a) The notion of total convexity at a point was first introduced in [13] but using the terminology ‘very convex’, and the notion of locally uniform total convexity at a point was introduced in [2]; while the notions of the essentially strict convexity, essentially smooth and Legendre were introduced in [10].

- (b) Clearly, the locally uniform total convexity at a point implies the total convexity at the same point. It was proved in [12] (see also [14, Proposition 2.2]) that if f is totally convex at any point of $\text{dom } f$, then it is strictly convex on $\text{dom } f$, and in [14, Proposition 2.13] that if X is reflexive and f is totally convex at any point of $\text{dom}(\partial f)$, then it is essentially strictly convex.

(c) By (7) and (8), the following equivalence holds:

$$J_X(x) \in \partial f^*(x^*) \iff x^* \in \partial f(x) \quad \text{for each pair } (x, x^*) \in X \times X^*; \quad (9)$$

hence, if f is Gâteaux differentiable on $\text{int}(\text{dom}f)$, then

$$J_X(x) \in (\partial f^* \circ \nabla f)(x) \quad \text{for each } x \in \text{int}(\text{dom}f)$$

and

$$x^* \in (\nabla f \circ J_X^{-1} \circ \partial f^*)(x^*) \quad \text{if } \partial f^*(x^*) \cap J_X(\text{int}(\text{dom}f)) \neq \emptyset.$$

(d) By [10, Theorem 5.6], if f is essentially smooth, then

$$\text{dom}(\partial f) = \text{int}(\text{dom}f); \quad (10)$$

hence, if f^* is also essentially smooth and if X is reflexive, then

$$\nabla f(\text{int}(\text{dom}f)) = \text{int}(\text{dom}f^*). \quad (11)$$

We end this section with two propositions on some properties of convex functions, which will be frequently used in subsequent sections; see [15, Proposition 2.11] and [16, Corollary 3.1, Corollary 3.2] for the first one, and [17, Proposition 3.4] for the second one.

Proposition 2.2: *Suppose that $f: X \rightarrow \overline{\mathbb{R}}$ is a proper lower semicontinuous convex function which is Gâteaux differentiable (resp., Fréchet differentiable) on $\text{int}(\text{dom}f)$. Then f is continuous and its Gâteaux derivative ∇f is norm-weak* continuous (resp., continuous) on $\text{int}(\text{dom}f)$.*

Proposition 2.3: *Let $x \in \text{dom}f$ and suppose that f is totally convex at x . Then $\partial f(x) \subseteq \text{int}(\text{dom}f^*)$ and f^* is Fréchet differentiable at each point $x^* \in \partial f(x)$. Furthermore, there exists a nondecreasing function $\theta: [0, +\infty) \rightarrow [0, +\infty)$ with $\lim_{t \rightarrow 0^+} \theta(t) = 0$ such that, for any pair $(y, y^*) \in X \times X^*$ with $y^* \in \partial f(y)$, one has*

$$\|y - x\| \leq \theta(\|y^* - x^*\|).$$

3. Chebyshev sets

Recall that X be a real Banach space, and that $f: X \rightarrow \overline{\mathbb{R}}$ be a proper lower semicontinuous convex function. Throughout the paper, we denote $U := \text{int}(\text{dom}f)$, $U^* := \text{int}(\text{dom}f^*)$, and we always assume that $U \neq \emptyset$, that C is a non-empty closed subset of U , and that f is Gâteaux differentiable on U with its Gâteaux

derivative denoted by ∇f . Recall that the Bregman distance D_f with respect to f is defined by (1). In particular, we have

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle \quad \text{for each pair } (x, y) \in X \times U. \quad (12)$$

Clearly, $D_f(\cdot, y)$ is convex on X for each $y \in U$. Moreover, for any $x, y, c \in U$, one has from (12) and (7) that

$$\begin{aligned} D_f(c, y) - D_f(x, y) &= f(c) - f(x) - \langle \nabla f(y), c - x \rangle, \\ D_f(x, y) &= D_f(x, c) - D_f(y, c) + \langle \nabla f(y) - \nabla f(c), y - x \rangle, \end{aligned} \quad (13)$$

and

$$D_f(x, y) = f(x) + f^*(\nabla f(y)) - \langle \nabla f(y), x \rangle. \quad (14)$$

The following result is useful, which can be shown by (9) (resp., Proposition 2.2).

Lemma 3.1: *Suppose that f and f^* are Gâteaux (resp., Fréchet) differential on U and $\nabla f(U)$, respectively. Then the mapping $\nabla f : U \rightarrow \nabla f(U)$ is a bijection (resp., isomorphism), and hence $(\nabla f)^{-1} = J_X^{-1} \circ \nabla f^*$ on $\nabla f(U)$.*

The following result was obtained in [1, Proposition 7.1] for the case of the Euclidean space $\mathbb{R}^n$, which can be used to study the properties of $\vec{\Pi}_C^f$ in terms of $\overleftarrow{\Pi}_{\nabla f(C)}^{f^*}$. Note that the right Bregman projection onto C can be defined similarly to (3) and denoted by $\vec{\Pi}_C^f$.

Lemma 3.2: *Suppose that f^* is Gâteaux differentiable on $\nabla f(U)$. Then the following assertions hold for each $x \in U$ and $y \in C$:*

$$y \in \vec{\Pi}_C^f(x) \iff \nabla f(y) \in \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(\nabla f(x)), \quad (15)$$

$$\vec{\Pi}_C^f(x) = J_X^{-1} \circ \nabla f^* \circ \overleftarrow{\Pi}_{\nabla f(C)}^{f^*} \circ \nabla f(x). \quad (16)$$

Proof: Let $x \in U$ and $y \in C$. Then, by Lemma 3.1, (14) and (8), we have

$$\begin{aligned} D_{f^*}(\nabla f(y), \nabla f(x)) &= f^*(\nabla f(y)) + f^{**}(\nabla f^*(\nabla f(x))) - \langle \nabla f^*(\nabla f(x)), \nabla f(y) \rangle \\ &= f^*(\nabla f(y)) + f(x) - \langle \nabla f(y), x \rangle \\ &= D_f(x, y). \end{aligned}$$

Hence,

$$\begin{aligned} y \in \vec{\Pi}_C^f(x) &\iff D_f(x, y) \leq D_f(x, c), \forall c \in C \\ &\iff D_{f^*}(\nabla f(y), \nabla f(x)) \leq D_{f^*}(\nabla f(c), \nabla f(x)), \forall c \in C \\ &\iff \nabla f(y) \in \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(\nabla f(x)), \end{aligned}$$

i.e. (15) holds. This and Lemma 3.1 imply assertion (16). ■

The following definition is taken from [2]; see also [1].

Definition 3.3: We say that C is

- (a) left Chebyshev (resp., left proximal) with respect to the left Bregman distance, or simply $\overleftarrow{D}_f$ -Chebyshev (resp., $\overleftarrow{D}_f$ -proximal) if for each $y \in U$, $\overleftarrow{\Pi}_C^f(y)$ is a singleton (resp., non-empty);
- (b) right Chebyshev (resp., right proximal) with respect to the right Bregman distance, or simply $\overrightarrow{D}_f$ -Chebyshev (resp., $\overrightarrow{D}_f$ -proximal) if for each $x \in U$, $\overrightarrow{\Pi}_C^f(x)$ is a singleton (resp., non-empty).

3.1. Left Bregman projections and left Chebyshev sets

For our study, we define the function $W^f : X \times X^* \rightarrow \overline{\mathbb{R}}$ associated with f by

$$W^f(x, x^*) := f(x) - \langle x^*, x \rangle + f^*(x^*) \quad \text{for each pair } (x, x^*) \in X \times X^*.$$

Clearly, W is nonnegative by (6), and the function $W(\cdot, x^*)$ is convex for any $x^* \in \text{dom } f^*$. Moreover, the following equality holds:

$$W^f(x, x^*) = W^f(x, y^*) + f^*(x^*) - f^*(y^*) - \langle x^* - y^*, x \rangle$$

for any $x^*, y^* \in X^*$ and $x \in X$. Similar to the case of Bregman distances, we define the relative distance function of C by

$$W_C^f(x^*) := \inf_{x \in C} W^f(x, x^*) \quad \text{for each } x^* \in \text{dom } f^*.$$

Then the relative projection operator onto C (relative to f) is defined by

$$P_C^f(x^*) := \{x \in C : W^f(x, x^*) = W_C^f(x^*)\} \quad \text{for each } x^* \in \text{dom } f^*.$$

The relationship between the two operators is described in the following lemma, which was given in [2, Proposition 3.1].

Lemma 3.4: *The following assertions hold:*

$$D_f(x, y) = W^f(x, \nabla f(y)) \quad \text{for each pair } (x, y) \in U \times U;$$

$$\overleftarrow{\Pi}_C^f(y) = P_C^f(\nabla f(y)) \quad \text{for each } y \in U. \quad (17)$$

Below we introduce the notion of a $\overleftarrow{D}_f$ -sun.

Definition 3.5: The set C is called

- (a) a $\overleftarrow{D}_f$ -sun if for each $x \in C$ and each $y \in U$,

$$x \in \overleftarrow{\Pi}_C^f(y) \implies x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$$

for any $\lambda \geq 0$ with $\overleftarrow{z}_\lambda^* \in \text{dom}(\partial f^*)$ and any $\overleftarrow{z}_\lambda \in J_X^{-1} \circ \partial f^*(\overleftarrow{z}_\lambda^*)$, where and in the sequel, $\overleftarrow{z}_\lambda^*$ is defined by

$$\overleftarrow{z}_\lambda^* := \lambda \nabla f(y) + (1 - \lambda) \nabla f(x); \quad (18)$$

- (b) a $\overleftarrow{D}_f$ -approximatively convex set if for each $y \in U$, $\overleftarrow{\Pi}_C^f(y)$ is non-empty convex.

The following Proposition 3.6(i) provides a characterization of a left Bregman nearest point, which corresponds to [1, Proposition 3.2 (i)] for the case of $\mathbb{R}^n$.

Proposition 3.6: Let $x \in C$ and $y \in U$. The following assertions hold.

- (i) $x \in \overleftarrow{\Pi}_C^f(y) \iff \langle \nabla f(y) - \nabla f(x), c - x \rangle \leq D_f(c, x)$ for each $c \in C$.
(ii) If $x \in \overleftarrow{\Pi}_C^f(y)$, then for each $\lambda \in [0, 1]$ such that $\overleftarrow{z}_\lambda^* \in \text{dom}(\partial f^*)$ and each $\overleftarrow{z}_\lambda \in J_X^{-1} \circ \partial f^*(\overleftarrow{z}_\lambda^*)$, one has that $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$.

Proof: Let $x \in C$ and $y \in U$. Assertion (i) was shown in [1, Proposition 3.2] for the case of $X = \mathbb{R}^n$, but its proof works here. Below we show that assertion (ii) holds. Let $x \in \overleftarrow{\Pi}_C^f(y)$ and $\lambda \in [0, 1]$ be such that $\overleftarrow{z}_\lambda^* \in \text{dom}(\partial f^*)$ and $\overleftarrow{z}_\lambda \in J_X^{-1} \circ \partial f^*(\overleftarrow{z}_\lambda^*)$. Since f is Gâteaux differentiable on U , it follows from (9) and (18) that

$$\nabla f(\overleftarrow{z}_\lambda) = \overleftarrow{z}_\lambda^* = \lambda \nabla f(y) + (1 - \lambda) \nabla f(x). \quad (19)$$

Hence,

$$\nabla f(\overleftarrow{z}_\lambda) - \nabla f(x) = \lambda [\nabla f(y) - \nabla f(x)], \quad (20)$$

and from (i), we have for each $c \in C$,

$$\langle \nabla f(\overleftarrow{z}_\lambda) - \nabla f(x), c - x \rangle = \lambda \langle \nabla f(y) - \nabla f(x), c - x \rangle \leq \lambda D_f(c, x) \leq D_f(c, x)$$

(noting that $D_f(c, x) \geq 0$ for all $c \in C$). This implies that $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$ again by assertion (i). The proof of (ii) is complete. $\blacksquare$

The following result shows that if f is strictly convex with $\nabla f(U) = X^*$, then a $\overleftarrow{D}_f$ -proximal set C is a $\overleftarrow{D}_f$ -sun if and only if it is convex.

Theorem 3.7: Assume that C is $\overleftarrow{D}_f$ -proximal.

(1) Consider the following statements:

- (i) C is convex.
- (ii) For each $x \in C$ and $y \in U$,

$$x \in \overleftarrow{\Pi}_C^f(y) \iff \langle \nabla f(y) - \nabla f(x), c - x \rangle \leq 0 \quad \text{for each } c \in C.$$

- (iii) C is a $\overleftarrow{D}_f$ -sun.

Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii).

(2) If f is strictly convex on U , then (i) $\iff$ (ii). Furthermore, if, in addition, $\nabla f(U) = X^*$, then (i) $\iff$ (ii) $\iff$ (iii), and each of them is equivalent to the following statement:

- (iv) For each $x \in C$ and $y \in U$, if $x \in \overleftarrow{\Pi}_C^f(y)$, then $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_2)$.

Proof: (1) (i) $\Rightarrow$ (ii) Suppose that (i) holds, and let $x \in C$ and $y \in U$. By Proposition 3.6(i), it suffices to show the ' $\Rightarrow$ ' part in (ii). To this end, let $x \in \overleftarrow{\Pi}_C^f(y)$. Then

$$\langle \nabla f(y) - \nabla f(x), c - x \rangle \leq D_f(c, x) \quad \text{for each } c \in C. \quad (21)$$

Let $c \in C$ be arbitrary. Since C is convex, one has that $\lambda c + (1 - \lambda)x \in C$ for all $\lambda \in (0, 1)$. Replacing c in (21) with $\lambda c + (1 - \lambda)x$, we obtain that

$$\begin{aligned} \lambda \langle \nabla f(y) - \nabla f(x), c - x \rangle &\leq D_f(\lambda c + (1 - \lambda)x, x) \\ &= f(\lambda c + (1 - \lambda)x) - f(x) - \lambda \langle \nabla f(x), c - x \rangle. \end{aligned}$$

It follows that

$$\begin{aligned} \langle \nabla f(y) - \nabla f(x), c - x \rangle &\leq \lim_{\lambda \rightarrow 0^+} \frac{f(\lambda c + (1 - \lambda)x) - f(x)}{\lambda} - \langle \nabla f(x), c - x \rangle \\ &= \langle \nabla f(x), c - x \rangle - \langle \nabla f(x), c - x \rangle \\ &= 0; \end{aligned}$$

hence, the ' $\Rightarrow$ ' part in (ii) holds.

(ii) $\Rightarrow$ (iii) Suppose that (ii) holds. Let $x \in C$ and $y \in U$ satisfy $x \in \overleftarrow{\Pi}_C^f(y)$. Let $\lambda \geq 0$ be such that $\overleftarrow{z}_\lambda^* \in \text{dom}(\partial f^*)$ and $\overleftarrow{z}_\lambda \in J_X^{-1} \circ \partial f^*(\overleftarrow{z}_\lambda^*)$. Then by (20) and (ii), one has that

$$\langle \nabla f(\overleftarrow{z}_\lambda) - \nabla f(x), c - x \rangle = \lambda \langle \nabla f(y) - \nabla f(x), c - x \rangle \leq 0 \quad \text{for each } c \in C;$$

hence, $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$ by (ii). By definition, C is a $\overleftarrow{D}_f$ -sun.

(2) Below we assume that f is strictly convex.

(ii) $\Rightarrow$ (i) Suppose that (ii) holds but (i) does not. Then there exist $c_1, c_2 \in C$ and $\lambda \in (0, 1)$ such that $y := \lambda c_1 + (1 - \lambda)c_2 \notin C$. Take $x \in \overleftarrow{\Pi}_C^f(y)$. Then

$$\langle \nabla f(y) - \nabla f(x), c_i - x \rangle \leq 0, \quad i = 1, 2$$

thanks to assumption. It follows that

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \leq 0.$$

Since ∇f is monotonic (see [9, Theorem 2.4.2 (iv), p.83]), one has that

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0;$$

hence,

$$\langle \nabla f(y) - \nabla f(x), y - x \rangle = 0.$$

This implies that

$$\begin{aligned} 0 \leq D_f(x, y) &= f(x) - f(y) - \langle \nabla f(y), x - y \rangle = f(x) - f(y) \\ &\quad + \langle \nabla f(x), y - x \rangle \leq 0; \end{aligned}$$

hence, $D_f(x, y) = 0$. Since f is strictly convex and since $x \neq y$ (noting that $y \notin C$), one has that

$$\begin{aligned} 0 \leq D_f\left(\frac{1}{2}x + \frac{1}{2}y, y\right) &= f\left(\frac{1}{2}x + \frac{1}{2}y\right) - f(y) - \left\langle \nabla f(y), \frac{1}{2}(x - y) \right\rangle \\ &< \frac{1}{2}(f(x) - f(y) - \langle \nabla f(y), x - y \rangle) = \frac{1}{2}D_f(x, y) = 0, \end{aligned}$$

which is a contradiction, and (i) is shown.

In the remainder of the proof of Theorem 3.7, we further assume that $\nabla f(U) = X^*$.

(iii) $\Rightarrow$ (ii) Suppose that (iii) holds. To show (ii), it suffices to show that the implication ' $\Rightarrow$ ' in (ii) holds. Let $x \in C$ and $y \in U$ satisfy $x \in \overleftarrow{\Pi}_C^f(y)$, and let $\lambda > 0$ be arbitrary. Since $\nabla f(U) = X^*$, $J_X^{-1} \circ \partial f^*(\overleftarrow{z}_\lambda^*) \neq \emptyset$, so $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$ for each $\overleftarrow{z}_\lambda \in J_X^{-1} \circ \partial f^*(\overleftarrow{z}_\lambda^*)$ because C is a $\overleftarrow{D}_f$ -sun. By Proposition 3.6(i) and (20), we have

$$\begin{aligned} \lambda \langle \nabla f(y) - \nabla f(x), c - x \rangle &= \langle \nabla f(\overleftarrow{z}_\lambda) - \nabla f(x), c - x \rangle \leq D_f(c, x) \\ &\text{for each } c \in C. \end{aligned}$$

Dividing two sides of the above inequality by λ and letting $\lambda \rightarrow +\infty$ yield

$$\langle \nabla f(y) - \nabla f(x), c - x \rangle \leq 0 \quad \text{for each } c \in C.$$

Hence, the implication ' $\Rightarrow$ ' of (ii) holds.

(iii) $\Rightarrow$ (iv) It is trivial.

(iv) $\Rightarrow$ (iii) Suppose that (iv) holds. Let $x \in C$ and $y \in U$. To show (iii), by Proposition 3.6(ii), it suffices to show that

$$x \in \overleftarrow{\Pi}_C^f(y) \implies x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda) \quad \text{for each } \lambda > 1 \text{ and each } \overleftarrow{z}_\lambda \in J_X^{-1} \circ \partial f^*(\overleftarrow{z}_\lambda^*) \quad (22)$$

satisfying $\overleftarrow{z}_\lambda^* \in \text{dom}(\partial f^*)$. To do this, let $x \in \overleftarrow{\Pi}_C^f(y)$, $\lambda > 1$, and $\overleftarrow{z}_\lambda$ be as above. Then, by induction, we have $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_{2^n})$; hence,

$$\nabla f(\overleftarrow{z}_{2^n}) = 2^n \nabla f(y) + (1 - 2^n) \nabla f(x), \quad n = 1, 2, \dots \quad (23)$$

by (19). Choose $n \in \mathbb{N}$ such that $2^n \geq \lambda$. Then $\frac{\lambda}{2^n} \leq 1$. By Proposition 3.6(ii), one has that

$$x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda) \text{ for all } \overleftarrow{z}_\lambda \in J_X^{-1} \circ \partial f^* \left(\frac{\lambda}{2^n} \nabla f(\overleftarrow{z}_{2^n}) + \left(1 - \frac{\lambda}{2^n}\right) \nabla f(x) \right). \quad (24)$$

Note from (23) that

$$\frac{\lambda}{2^n} \nabla f(\overleftarrow{z}_{2^n}) + \left(1 - \frac{\lambda}{2^n}\right) \nabla f(x) = \lambda \nabla f(y) + (1 - \lambda) \nabla f(x).$$

Hence,

$$\partial f^* \left(\frac{\lambda}{2^n} \nabla f(\overleftarrow{z}_{2^n}) + \left(1 - \frac{\lambda}{2^n}\right) \nabla f(x) \right) = \partial f^*(\lambda \nabla f(y) + (1 - \lambda) \nabla f(x)),$$

and consequently, $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$ by (24), which completes the proof of (22). $\blacksquare$

Remark 3.1: If f is not strictly convex on U , then the implication (ii) $\Rightarrow$ (i) does hold in general. In fact, let $X = \mathbb{R}$ and define the function f by $f(x) = x$ for each $x \in \mathbb{R}$. Then $U = \mathbb{R}$, and $\nabla f \equiv 1$ on $\mathbb{R}$. Furthermore, Let $C = [-2, -1] \cup [1, 2]$. Clearly, C is $\overleftarrow{D}_f$ -proximal, and condition (ii) holds in Theorem 3.7, but C is not convex.

Recall that $S \subseteq X$ is boundedly compact if for any $x \in S$ and any $\delta > 0$, $S \cap \mathbf{B}(x, \delta)$ is compact. It is easy to see that if S is boundedly compact, then S is closed.

By [2, Theorem 3.2 and Proposition 3.3], we have the following useful result.

Lemma 3.8: Suppose that C be a boundedly compact, and that f is locally uniformly totally convex at any point of U .

(1) $\overleftarrow{\Pi}_C^f(y) \neq \emptyset$ for each $y \in U$.

(2) If C is $\overleftarrow{D}_f$ -Chebyshev, then

(i) the operator $P_C^f(\cdot)$ is continuous on $\nabla f(U)$;

- (ii) the operator $\overleftarrow{\Pi}_C^f(\cdot)$ is continuous on U provided that f is additionally Fréchet differential on U .

The following result illustrates that if f has sufficiently good properties, then for a boundedly compact set C , the convexity, the Chebyshev property, and the $\overleftarrow{D}_f$ -solarity of C are equivalent mutually.

Theorem 3.9: Suppose that f is locally uniformly totally convex and Fréchet differential at any point of U , and that $\nabla f(U) = U^*$. Let C be a boundedly compact set. Consider the following statements:

- (i) C is convex.
- (ii) C is $\overleftarrow{D}_f$ -approximatively convex.
- (iii) C is $\overleftarrow{D}_f$ -Chebyshev.
- (iv) C is a $\overleftarrow{D}_f$ -sun of U .

Then (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv). If, in addition, $\nabla f(U) = X^*$, then all statements above are equivalent mutually.

Proof: (i) $\Rightarrow$ (ii) Suppose that (i) holds and let $y \in U$. Then $\overleftarrow{\Pi}_C^f(y) \neq \emptyset$ by Lemma 3.8(1). To show $\overleftarrow{\Pi}_C^f(y)$ is convex, let $x_1, x_2 \in \overleftarrow{\Pi}_C^f(y)$, and let $\lambda \in [0, 1]$. Then, $D_f(x_1, y) = D_f(x_2, y) = \overleftarrow{D}_C^f(y)$, and $\lambda x_1 + (1 - \lambda)x_2 \in C$. It follows that

$$\overleftarrow{D}_C^f(y) \leq D_f(\lambda x_1 + (1 - \lambda)x_2, y) \leq \lambda D_f(x_1, y) + (1 - \lambda)D_f(x_2, y) = \overleftarrow{D}_C^f(y)$$

because $D_f(\cdot, y)$ is convex on U ; hence, $\lambda x_1 + (1 - \lambda)x_2 \in \overleftarrow{D}_C^f(y)$. Therefore, C is $\overleftarrow{D}_f$ -approximatively convex.

(ii) $\Rightarrow$ (iii) Suppose that (ii) holds, and let $y \in U$. Then $\overleftarrow{\Pi}_C^f(y) \neq \emptyset$. If there exist $x_1, x_2 \in \overleftarrow{\Pi}_C^f(y)$, $x_1 \neq x_2$, then $(x_1 + x_2)/2 \in \overleftarrow{\Pi}_C^f(y)$ by (ii). Since f is locally uniformly totally convex at any point of U , f is strictly convex on U by Remark 2.2(b), so is $D_f(\cdot, y)$. Thus,

$$\overleftarrow{D}_C^f(y) = D_f\left(\frac{1}{2}x_1 + \frac{1}{2}x_2, y\right) < \frac{1}{2}D_f(x_1, y) + \frac{1}{2}D_f(x_2, y) = \overleftarrow{D}_C^f(y),$$

which is a contradiction and (iii) holds.

(iii) $\Rightarrow$ (iv) Suppose that (iii) holds but C is not a $\overleftarrow{D}_f$ -sun of U . Then there exist $x \in C$ and $y \in U$ such that $x \in \overleftarrow{\Pi}_C^f(y)$ but $x \notin \overleftarrow{\Pi}_C^f(\overleftarrow{z}_{\lambda_1})$ for some $\lambda_1 \geq 0$ and some

$$\overleftarrow{z}_{\lambda_1} \in J_X^{-1} \circ \partial f^*(\lambda_1 \nabla f(y) + (1 - \lambda_1) \nabla f(x)). \quad (25)$$

Since f^* is Fréchet differential on $\nabla f(U)$ by hypothesis and Proposition 2.3, one has that

$$\overleftarrow{z}_{\lambda} = J_X^{-1} \circ \partial f^*(\lambda \nabla f(y) + (1 - \lambda) \nabla f(x)) \quad \text{for all } \lambda \geq 0. \quad (26)$$

We assert that $y \notin C$. Otherwise, since $D_f(y, y) = 0$, one has that $y = \overleftarrow{\Pi}_C^f(y) = x$. Hence, thanks to (25) and (26), $\overleftarrow{z}_{\lambda_1} = J_X^{-1} \circ \partial f^* \circ \nabla f(x) = x$. So, $x \in \overleftarrow{\Pi}_C^f(x) = \overleftarrow{\Pi}_C^f(\overleftarrow{z}_{\lambda_1})$, which is a contradiction, and the proof of the assertion is complete. Since for each $\lambda \in [0, 1]$, $\overleftarrow{z}_\lambda^* = \lambda \nabla f(y) + (1 - \lambda) \nabla f(x) \in U^* = \nabla f(U)$ (noting that U^* is convex), one has that $\overleftarrow{z}_\lambda^* \in \text{dom}(\partial f^*)$. This together with Proposition 3.6(ii) implies that $x \in \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$. Therefore, $\lambda_1 > 1$.

Now let $\lambda_0 := \sup\{\lambda > 0 : x = \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)\}$. Then $\lambda_0 \geq 1$. Without loss of generality, we may assume that $\lambda_0 = 1$. Then $x = \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda)$ for each $\lambda \in [0, 1]$, but

$$x \neq \overleftarrow{\Pi}_C^f(\overleftarrow{z}_\lambda) \quad \text{for all } \lambda > 1. \quad (27)$$

Since C is boundedly compact, we may take $\epsilon_0 > 0$ such that $C \cap \mathbf{B}(x, \epsilon_0)$ is compact. Note from Lemma 3.4 that $x = \overleftarrow{\Pi}_C^f(y) = P_C^f(\nabla f(y))$. Note also from Lemma 3.8(2) that P_C^f is continuous at $\nabla f(y)$. There exists $\delta > 0$ such that

$$\mathbf{B}(\nabla f(y), \delta) \subseteq U^* = \nabla f(U) \quad \text{and} \quad P_C^f(\mathbf{B}(\nabla f(y), \delta)) \subseteq C \cap \mathbf{B}(x, \epsilon_0). \quad (28)$$

Since $\nabla f : U \rightarrow \nabla f(U)$ is an isomorphism by Lemma 3.1, $\nabla f(y) \notin \nabla f(C)$ as $y \notin C$.

Now define maps $\phi : \nabla f(C) \rightarrow \mathbf{B}^* := \mathbf{B}(\nabla f(y), \delta)$ by

$$\phi(c^*) := \nabla f(y) + \delta \frac{\nabla f(y) - c^*}{\|\nabla f(y) - c^*\|} \quad \text{for each } c^* \in \nabla f(C) \quad (29)$$

and $T : \mathbf{B}^* \rightarrow \mathbf{B}^*$ by

$$T := \phi \circ \nabla f \circ P_C^f.$$

Then T is continuous, and

$$T(\mathbf{B}^*) = \phi \circ \nabla f \circ P_C^f(\mathbf{B}^*) \subseteq \phi \circ \nabla f(C \cap \mathbf{B}(x, \epsilon_0)) \subseteq \mathbf{B}^*.$$

This implies that $T(\mathbf{B}^*)$ is relative compact because $\phi \circ \nabla f(C \cap \mathbf{B}(x, \epsilon_0))$ is compact.

Thus, the Schauder fixed theorem (see, for example [18, Theorem 8.8, p.60]) is applicable and there exists $z^* \in \mathbf{B}^*$ such that $T(z^*) = z^*$. Since $z^* \in \nabla f(U)$ by (28), there is $z \in U$ such that $z^* = \nabla f(z)$. Thus

$$T(z^*) = z^* = \nabla f(z), \quad (30)$$

and

$$T(z^*) = \phi \circ \nabla f \circ P_C^f(z^*) = \phi \circ \nabla f \circ P_C^f(\nabla f(z)) = \phi \circ \nabla f \circ \overleftarrow{\Pi}_C^f(z)$$

thanks to (17). It follows from (29) and (30) that

$$\nabla f(z) = \phi \circ \nabla f \circ \overleftarrow{\Pi}_C^f(z) = \nabla f(y) + \delta \frac{\nabla f(y) - \nabla f(\overleftarrow{\Pi}_C^f(z))}{\|\nabla f(y) - \nabla f(\overleftarrow{\Pi}_C^f(z))\|}.$$

Let $t := \delta / \|\nabla f(y) - \nabla f(\overleftarrow{\Pi}_C^f(z))\| > 0$. Then

$$\nabla f(z) = (1+t)\nabla f(y) - t\nabla f(\overleftarrow{\Pi}_C^f(z)). \quad (31)$$

Hence,

$$\nabla f(y) = \frac{1}{1+t}\nabla f(z) + \frac{t}{1+t}\nabla f(\overleftarrow{\Pi}_C^f(z)),$$

and so

$$y = J_X^{-1} \circ \nabla f^* \left[\frac{1}{1+t}\nabla f(z) + \frac{t}{1+t}\nabla f(\overleftarrow{\Pi}_C^f(z)) \right].$$

Applying Proposition 3.6(ii) (to z , $\overleftarrow{\Pi}_C^f(z)$ and $1/(1+t)$ in place of y , x and λ) and noting that C is a $\overleftarrow{D}_f$ -Chebyshev set, one has that

$$\overleftarrow{\Pi}_C^f(y) = \overleftarrow{\Pi}_C^f(z) = x.$$

This and (31) imply that

$$\overleftarrow{z}_{1+t}^* = \nabla f(z) = (1+t)\nabla f(y) + (1-(1+t))\nabla f(x).$$

It then follows from Lemma 3.1 that $\overleftarrow{z}_{1+t} = (\nabla f)^{-1}(\overleftarrow{z}_{1+t}^*) = (\nabla f)^{-1}(\nabla f(z)) = z$. Hence, $x = \overleftarrow{\Pi}_C^f(z) = \overleftarrow{\Pi}_C^f(\overleftarrow{z}_{1+t})$, which contradicts (27) and the proof of (iii) $\Rightarrow$ (iv) is complete.

In the rest of the proof of Theorem 3.9, we assume that $\nabla f(U) = X^*$. Thus to complete the proof of Theorem 3.9, it suffices to show (iv) $\Rightarrow$ (i), which is clear by the equivalence of (i) and (iii) in Theorem 3.7. $\blacksquare$

Remark 3.2: The implication (iii) $\Rightarrow$ (iv) in Theorem 3.9 corresponds to a classical result in the approximation theory based on norms, which asserts that each bounded compact Chebyshev set in Banach spaces is a sun; see [19] or [8, Theorem 1.1, p.149]. A similar result for the case of the right Bregman distance sees Theorem 3.13 below.

3.2. Right Bregman projections and right Chebyshev sets

Recall that the right Chebyshev set was defined by Definitions 3.3. Then the right Bregman nearest point is characterized as follows, which is similar to [1, Proposition 3.2(i)].

Proposition 3.10: Let $x \in U$ and $y \in C$. Then the following assertions hold.

- (i) $y \in \overrightarrow{\Pi}_C^f(x) \iff \langle \nabla f(y) - \nabla f(c), y - x \rangle \leq D_f(y, c)$ for each $c \in C$.
(ii) If $y \in \overrightarrow{\Pi}_C^f(x)$, then for each $\lambda \in [0, 1]$, $y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda)$, where
- $$\overrightarrow{z}_\lambda := \lambda x + (1 - \lambda)y. \quad (32)$$

Proof: (i) By the definitions of the Bregman distance and the right Bregman projection,

$$\begin{aligned} y \in \overrightarrow{\Pi}_C^f(x) &\iff D_f(x, c) - D_f(x, y) \geq 0, \quad \forall c \in C \\ &\iff f(y) - f(c) - \langle \nabla f(c), y - c \rangle + \langle \nabla f(y) - \nabla f(c), x - y \rangle \geq 0, \quad \forall c \in C \\ &\iff D_f(y, c) \geq \langle \nabla f(y) - \nabla f(c), y - x \rangle, \quad \forall c \in C. \end{aligned}$$

- (ii) Let $y \in \overrightarrow{\Pi}_C^f(x)$ and $\lambda \in [0, 1]$. Note that

$$y - \overrightarrow{z}_\lambda = \lambda(y - x). \quad (33)$$

Then, by (i), we have that

$$\begin{aligned} \langle \nabla f(y) - \nabla f(c), y - \overrightarrow{z}_\lambda \rangle &= \lambda \langle \nabla f(y) - \nabla f(c), y - x \rangle \leq D_f(y, c) \\ &\text{for each } c \in C; \end{aligned}$$

hence, $y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda)$. The proof is complete. ■

Below we introduce the definition of $\overrightarrow{D}_f$ -suns.

Definition 3.11: We say that C is a $\overrightarrow{D}_f$ -sun if, for each $x \in U$ and each $y \in C$,

$$y \in \overrightarrow{\Pi}_C^f(x) \implies y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda) \quad \text{for each } \lambda \geq 0 \text{ with } \overrightarrow{z}_\lambda \in U,$$

where $\overrightarrow{z}_\lambda$ was defined by (32).

The following result shows that if $U = X$, $\nabla f(U) = U^*$, and f^* is Gâteaux differential and strictly convex on U^* , then a $\overrightarrow{D}_f$ -proximal set C is a $\overrightarrow{D}_f$ -sun if and only if $\nabla f(C)$ is convex.

Theorem 3.12: Suppose that C is $\overrightarrow{D}_f$ -proximal.

- (1) In the following statements, (i) $\implies$ (ii).

- (i) For each $x \in U$ and $y \in C$,

$$y \in \overrightarrow{\Pi}_C^f(x) \iff \langle \nabla f(y) - \nabla f(c), y - x \rangle \leq 0 \quad \text{for each } c \in C. \quad (34)$$

- (ii) C is a $\overrightarrow{D}_f$ -sun.
- (2) If $U=X$, then (i) $\iff$ (ii), and each of them is equivalent to the following statement:
- (iii) For each $x \in U$ and $y \in C$, if $y \in \overrightarrow{\Pi}_C^f(x)$, then $y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_2)$.
- (3) If $\nabla f(U) = U^*$, and if f^* is Gâteaux differential and strictly convex on U^* , then (i) is equivalent to the following statement:
- (iv) $\nabla f(C)$ is convex;

consequently, if, in addition, $U=X$, then each of above all statements is equivalent mutually.

Proof: (1) (i) $\implies$ (ii) Suppose that (i) holds. Let $x \in U$, $y \in \overrightarrow{\Pi}_C^f(x)$, and $\lambda \geq 0$ be such that $\overrightarrow{z}_\lambda \in U$. Then by (33) and (i), one has that

$$\langle \nabla f(y) - \nabla f(c), y - \overrightarrow{z}_\lambda \rangle = \lambda \langle \nabla f(y) - \nabla f(c), y - x \rangle \leq 0 \leq D_f(y, c)$$

for all $c \in C$,

which implies that $y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda)$ thanks to Proposition 3.10(i). Hence, C is a $\overrightarrow{D}_f$ -sun.

(2) Below in the proofs of (ii) $\implies$ (i) and (ii) $\iff$ (iii), we assume that $U=X$.

(ii) $\implies$ (i) Suppose that (ii) holds. Let $x \in U$ and $y \in C$. To show (i), it suffices to show that the implication ' $\implies$ ' in (34) holds. To this end, let $y \in \overrightarrow{\Pi}_C^f(x)$. Then, for each $\lambda > 0$, $y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda)$. Hence,

$$\lambda \langle \nabla f(y) - \nabla f(c), y - x \rangle = \langle \nabla f(y) - \nabla f(c), y - \overrightarrow{z}_\lambda \rangle \leq D_f(y, c)$$

for each $c \in C$.

Dividing two sides of the above inequality by λ and letting $\lambda \rightarrow \infty$ yield

$$\langle \nabla f(y) - \nabla f(c), y - x \rangle \leq 0 \quad \text{for each } c \in C,$$

that is, the implication ' $\implies$ ' in (34) holds.

(ii) $\implies$ (iii) It is trivial.

(iii) $\implies$ (ii) Suppose that (iii) holds. To show (ii), it suffices to show that for each $x \in U$ and each $y \in C$,

$$y \in \overrightarrow{\Pi}_C^f(x) \implies y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda) \quad \text{for each } \lambda > 1. \quad (35)$$

To do this, Let $x \in U$, $y \in \overrightarrow{\Pi}_C^f(x)$, and $\lambda > 1$. Then, by induction, we have $y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_{2^n})$. Choose $n \in \mathbb{N}$ such that $2^n \geq \lambda$. Then $\frac{\lambda}{2^n} \leq 1$. By Proposition 3.10(ii), one has that

$$y \in \overrightarrow{\Pi}_C^f\left(\frac{\lambda}{2^n} \overrightarrow{z}_{2^n} + \left(1 - \frac{\lambda}{2^n}\right)y\right) = \overrightarrow{\Pi}_C^f(\lambda x + (1 - \lambda)y) = \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda).$$

Hence, (35) holds.

(3) Suppose that $\nabla f(U) = U^*$, and that f^* is Gâteaux differential and strictly convex on U^* . Then $\nabla f(C)$ is $\overleftarrow{D}_{f^*}$ -proximal in U^* . Indeed, let $x^* \in U^*$. Then there is $x \in U$ such that $\nabla f(x) = x^*$. Thus, $\overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(x^*) = \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(\nabla f(x)) \neq \emptyset$ by (15) because $\overrightarrow{\Pi}_C^f(x) \neq \emptyset$ by hypothesis; hence, $\nabla f(C)$ is $\overleftarrow{D}_{f^*}$ -proximal because $x^* \in U^*$ is arbitrary. Below we show that (i) $\iff$ (iv). By (15), we see that (i) holds if and only if the following assertion holds:

(i') For each $x \in U$ and $y \in C$,

$$\nabla f(y) \in \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(\nabla f(x)) \iff \langle \nabla f(y) - \nabla f(c), y - x \rangle \leq 0 \quad \text{for each } c \in C.$$

Since for $x \in U, y, c \in C$,

$$\langle \nabla f(y) - \nabla f(c), y - x \rangle = \langle \nabla f^*(\nabla f(x)) - \nabla f^*(\nabla f(y)), \nabla f(c) - \nabla f(y) \rangle$$

by Lemma 3.1, and since $\nabla f(U) = U^*$, one has that (i') holds if and only if the following assertion holds:

(i'') For each $x^* \in U^*$ and $y^* \in \nabla f(C)$,

$$y^* \in \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(x^*) \iff \langle \nabla f^*(x^*) - \nabla f^*(y^*), c^* - y^* \rangle \leq 0 \quad \text{for each } c^* \in \nabla f(C).$$

Thus by the equivalence (i) $\iff$ (ii) in Theorem 3.7 (2) (noting that f^* is strictly convex on U^*), (i'') $\iff \nabla f(C)$ is convex. Therefore, (i) $\iff$ (iv).

Finally, we further assume that $U = X$, it is easy to see that all statements in Theorem 3.12 are equivalent each other. The proof of Theorem 3.12 is complete. ■

The following result provides a sufficient condition for a boundedly compact set to be a $\overrightarrow{D}_f$ -sun.

Theorem 3.13: *Suppose that f is totally convex at any point of U , that $\nabla f(U) = U^*$, and that f^* is locally uniformly totally convex at any point of U^* . If C be a boundedly compact $\overrightarrow{D}_f$ -Chebyshev set, then C is a $\overrightarrow{D}_f$ -sun.*

Proof: Suppose that the condition of Theorem 3.13 holds. We first show that $\nabla f(C)$ is $\overrightarrow{D}_{f^*}$ -Chebyshev in U^* . In fact, since f is totally convex at any point of U , it follows from Proposition 2.3 that f^* is Fréchet (hence Gâteaux) differentiable on $\nabla f(U)$. By (15), $\nabla f(C)$ is $\overleftarrow{D}_{f^*}$ -Chebyshev because C is $\overrightarrow{D}_f$ -Chebyshev.

We then show that $\nabla f(C)$ is bounded compact. Indeed, note that f^{**} is Fréchet differentiable on $\partial f^*(U^*)$ by Remark 2.2(b) and Proposition 2.3. Note also that $f^{**} \circ J_X = f$ on X and $J_X^{-1} \circ \partial f^*(U^*) = J_X^{-1} \circ \nabla f^*(U^*) = U$. One has that f is Fréchet differentiable on U . Hence, ∇f is continuous on U by Proposition 2.2. To proceed, let $x^* \in \nabla f(C)$ and $\delta > 0$ are arbitrary, and let $\{y_n^*\} \subseteq \nabla f(C) \cap \mathbf{B}(x^*, \delta)$.

Then for each $n \in \mathbb{N}$ there is $y_n \in C$ such that $\nabla f(y_n) = y_n^*$. Applying Proposition 2.3 (to y_1 and y_1^* in place of x and $x^* \in \partial f(x)$), there exists a nondecreasing function $\theta : [0, +\infty) \rightarrow [0, +\infty)$ with $\lim_{t \rightarrow 0^+} \theta(t) = 0$ such that

$$\|y_n - y_1\| \leq \theta(\|\nabla f(y_n) - \nabla f(y_1)\|) \quad \text{for all } n \in \mathbb{N}.$$

Noting that

$$\|\nabla f(y_n) - \nabla f(y_1)\| = \|y_n^* - y_1^*\| \leq \|y_n^* - x^*\| + \|x^* - y_1^*\| \leq 2\delta$$

by the choice of $\{y_n^*\}$, one has that $\|y_n - y_1\| \leq \theta(2\delta)$ for all $n \in \mathbb{N}$. This show that $\{y_n\}$ is bounded. Since $C \subseteq U$ is bounded compact, there exists a subsequence $\{y_{n_k}\}$ of $\{y_n\}$ such that $\lim_{k \rightarrow \infty} y_{n_k} = y$ for some $y \in C$. Thus, $\lim_{k \rightarrow \infty} y_{n_k}^* = \lim_{k \rightarrow \infty} \nabla f(y_{n_k}) = \nabla f(y) \in \nabla f(C) \cap \mathbf{B}(x^*, \delta)$. Therefore, $\nabla f(C)$ is bounded compact.

Thirdly, we show that $\nabla f(C)$ is a $\overleftarrow{D}_{f^*}$ -sun in U^* . In fact, since f^* is locally uniformly totally convex at any point of U^* and Fréchet differentiable on $\nabla f(U) = U^*$, a similar proof to Theorem 3.9 (iii) $\Rightarrow$ (iv), with f^* , $\nabla f(C)$, and U^* in place of f , C , and U , yields that $\nabla f(C)$ is a $\overleftarrow{D}_{f^*}$ -sun in U^* .

Finally, we show that C is a $\overrightarrow{D}_f$ -sun. To this end, let $x \in U$ and $y \in C$ satisfy $y = \overrightarrow{\Pi}_C^f(x)$ (noting that C is a $\overrightarrow{D}_f$ -Chebyshev set by hypothesis). One has to show that $y \in \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda)$ for all $\lambda \geq 0$ with $\overrightarrow{z}_\lambda \in U$. Now fix $\lambda \geq 0$ with $\overrightarrow{z}_\lambda \in U$. Since $y = \overrightarrow{\Pi}_C^f(x)$, one obtains from Lemma 3.2 that $\nabla f(y) = \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(\nabla f(x))$. Since $\nabla f(C)$ is a $\overleftarrow{D}_{f^*}$ -Chebyshev $\overleftarrow{D}_{f^*}$ -sun in U^* , one has from the definition of $\overleftarrow{D}_{f^*}$ -sun and Lemma 3.1 that

$$\begin{aligned} \nabla f(y) &= \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(J_X^{-1} \circ \partial f^{**}(\lambda \nabla f^* \circ \nabla f(x)) + (1 - \lambda) \nabla f^* \circ \nabla f(y)) \\ &= \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}((\nabla f^*)^{-1} \circ J_X(\lambda x + (1 - \lambda)y)) \\ &= \overleftarrow{\Pi}_{\nabla f(C)}^{f^*}(\nabla f(\lambda x + (1 - \lambda)y)) \end{aligned}$$

(noting that f^{**} is Fréchet differentiable on $\nabla f^*(U^*) = J_X(U)$). This, together with Lemma 3.1 and (16), implies that

$$y = J_X^{-1} \circ \nabla f^* \circ \overleftarrow{\Pi}_{\nabla f(C)}^{f^*} \circ \nabla f(\lambda x + (1 - \lambda)y) = \overrightarrow{\Pi}_C^f(\overrightarrow{z}_\lambda).$$

Hence, C is a $\overrightarrow{D}_f$ -sun in U and the proof is complete. ■

By Theorems 3.12 and 3.13, Proposition 2.3 and Remark 2.2(b), one has the following result.

Corollary 3.14: *Suppose that f is totally convex at any point of X , $\nabla f(X) = U^*$, and that f^* is locally uniformly totally convex at any point of U^* . If C be a boundedly compact $\overrightarrow{D}_f$ -Chebyshev set, then $\nabla f(C)$ is convex.*

Remark 3.3: From Theorems 3.9 and 3.13 (and their proofs), we see that the function f there and its dual f^* are required to be Fréchet differentiable on U and $\nabla f(U)$, respectively. Below we provide two such examples.

Example 3.15: Let $1 < p < +\infty$ and let l_p denote the space of all p -powers summable sequences of real numbers with the usual l_p -norm $\|\cdot\|_p$. Define the function $f = \frac{1}{p}\|\cdot\|_p^p$ on l_p . Then $f^* = \frac{1}{q}\|\cdot\|_q^q$ by [10, Lemma 6.2], where $\frac{1}{p} + \frac{1}{q} = 1$. Clearly, f and f^* are, respectively, Fréchet differentiable on l_p and l_q because the norms of l_p and l_q uniformly Fréchet differentiable.

Example 3.16: (A revision of [10, Example 6.3].) Let X be a reflexive strictly convex Banach space with the Fréchet differentiable norm and the Fréchet differentiable dual norm. Define

$$f(x) = \begin{cases} -\sqrt{1 - \|x\|^2} & \text{if } \|x\| \leq 1; \\ +\infty & \text{otherwise.} \end{cases}$$

Then $\text{dom } f = \mathbf{B}(0, 1)$ and $f^*(x^*) = \sqrt{1 + \|x^*\|^2}$ for all $x^* \in X^*$. Moreover,

$$\nabla f(x) = \frac{\|x\|\nabla\|x\|}{\sqrt{1 - \|x\|^2}} \text{ for } x \in U \quad \text{and} \quad \nabla f^*(x^*) = \frac{\|x^*\|\nabla\|x^*\|}{\sqrt{1 + \|x^*\|^2}} \text{ for } x^* \in X^*.$$

It is not hard to see that f and f^* are Fréchet differentiable on U and $\nabla f(U) = X^*$ (here, $\nabla f(U) = X^*$ is because of [10, Theorem 5.10]), respectively.

4. Bregman farthest-point maps and Klee sets

In this section, we assume that C is a non-empty bounded closed subset of U . Recall that the left and right Bregman farthest distance function of C and the left Bregman farthest-point map onto C are defined by (2) and (5) and (4), respectively. Similarly, we can define the right Bregman farthest-point map onto C and denote it by $\vec{Q}_C^f$.

Definition 4.1: We say that the set C is right Klee with respect to the right Bregman distance, or simply $\vec{D}_f$ -Klee, if for each $x \in U$, $\vec{Q}_C^f(x)$ is a singleton.

Repeating the proof of [3, Proposition 7.1] yields the following useful lemma.

Lemma 4.2: Suppose that f^* is Gâteaux differential on U^* and that $\nabla f(U) = U^*$. Then

$$\vec{Q}_C^f = J_X^{-1} \circ \nabla f^* \circ \overleftarrow{Q}_{\nabla f(C)}^{f^*} \circ \nabla f. \quad (36)$$

Lemma 4.3: Suppose that C is compact with $\overline{\text{co } C} \subseteq U$. Then the following assertions hold.

- (i) $\max_{x \in \overline{\text{co } C}} f(x) = \max_{x \in C} f(x)$.
- (ii) If f is strictly convex on U , then

$$f(c) < \max_{x \in C} f(x) \quad \text{for each } c \in \overline{\text{co } C} \setminus \text{ext } C.$$

Proof: Since C is compact, it follows from [20, Theorem, p.61] that $\overline{\text{co } C}$ is compact. Thus, assertion (i) is seen to hold. To show assertion (ii), we suppose, on the contrary, that there is $c_0 \in \overline{\text{co } C} \setminus \text{ext } C$ such that $f(c_0) = \max_{x \in C} f(x)$. Then there exist $c_1, c_2 \in C$, $\lambda_0 \in (0, 1)$ such that $c_0 = \lambda_0 c_1 + (1 - \lambda_0) c_2$. It follows from the strict convexity of f that

$$\max_{x \in C} f(x) = f(c_0) < \lambda_0 f(c_1) + (1 - \lambda_0) f(c_2) \leq \max_{x \in C} f(x),$$

which is a contradiction, and the proof of (ii) is complete. ■

Proposition 4.4: Suppose that f is Legendre and that C is a compact $\overleftarrow{D}_f$ -Klee set with $\overline{\text{co } C} \subseteq U$. Then $\overline{\text{co } C}$ is a compact $\overleftarrow{D}_f$ -Klee set.

Proof: As noted earlier, $\overline{\text{co } C}$ is compact. To show $\overline{\text{co } C}$ is $\overleftarrow{D}_f$ -Klee, let $y \in U$. Since f is Legendre, it is essentially strict and essentially smooth. Thus, by Remark 2.2(d), $\text{dom}(\partial f) = U$ is convex. Hence, f is strictly convex on U , and so is the function $D_f(\cdot, y)$ on U . Since C is a compact $\overleftarrow{D}_f$ -Klee set, one has from (4) and Lemma 4.3(i) that

$$D_f(\overleftarrow{Q}_C^f(y), y) = \overleftarrow{F}_C^f(y) = \overleftarrow{F}_{\overline{\text{co } C}}^f(y), \quad (37)$$

and for each $c \in C \setminus \{\overleftarrow{Q}_C^f(y)\}$,

$$D_f(c, y) < \overleftarrow{F}_C^f(y). \quad (38)$$

Moreover, for each $c \in \overline{\text{co } C} \setminus C$, (38) remains to hold by Lemma 4.3(ii) because $\overline{\text{co } C} \setminus C \subseteq \overline{\text{co } C} \setminus \text{ext } C$. Noting that $\overline{\text{co } C} \setminus \{\overleftarrow{Q}_C^f(y)\} = (\overline{\text{co } C} \setminus C) \cup (C \setminus \{\overleftarrow{Q}_C^f(y)\})$, we see that (38) holds for each $c \in \overline{\text{co } C} \setminus \{\overleftarrow{Q}_C^f(y)\}$. This together with (37) implies that $\overleftarrow{Q}_{\overline{\text{co } C}}^f(y) = \{\overleftarrow{Q}_C^f(y)\}$; hence, $\overline{\text{co } C}$ is a $\overleftarrow{D}_f$ -Klee set. The proof is complete. ■

Lemma 4.5: Suppose that C is compact. Then $\overleftarrow{F}_C^f(\cdot)$ is continuous on U . Furthermore, if C is additionally $\overleftarrow{D}_f$ -Klee, then $\overleftarrow{Q}_C^f(\cdot)$ is continuous on U .

Proof: Let $y_0 \in U$ be arbitrary and let $\{y_n\}$ be a sequence in U convergent to y_0 . Since ∇f is locally bounded at y_0 , $\{\nabla f(y_n)\}$ is bounded. Thus, by the compactness of C and the norm-weak* continuity of ∇f at y_0 , it is not difficult to show that

$$\lim_{n \rightarrow \infty} \sup_{x \in C} \langle \nabla f(y_n) - \nabla f(y_0), y_0 - x \rangle = 0. \quad (39)$$

Note from (13) that

$$D_f(x, y_n) = D_f(x, y_0) - D_f(y_n, y_0) + \langle \nabla f(y_n) - \nabla f(y_0), y_n - x \rangle.$$

Taking supremums in two sides of the above equality with respect to $x \in C$ yields

$$\begin{aligned} \overleftarrow{F}_C^f(y_n) &\leq \overleftarrow{F}_C^f(y_0) - D_f(y_n, y_0) + \langle \nabla f(y_n) - \nabla f(y_0), y_n - y_0 \rangle \\ &\quad + \sup_{x \in C} \langle \nabla f(y_n) - \nabla f(y_0), y_0 - x \rangle. \end{aligned} \quad (40)$$

Note that $\lim_{n \rightarrow \infty} D_f(y_n, y_0) = 0$ and that

$$\lim_{n \rightarrow \infty} \langle \nabla f(y_n) - \nabla f(y_0), y_n - y_0 \rangle = 0.$$

We obtain from (39) and (40) that

$$\overline{\lim}_{n \rightarrow \infty} \overleftarrow{F}_C^f(y_n) \leq \overleftarrow{F}_C^f(y_0). \quad (41)$$

On the other hand, since

$$D_f(x, y_0) = D_f(x, y_n) - D_f(y_0, y_n) + \langle \nabla f(y_0) - \nabla f(y_n), y_0 - x \rangle$$

again by (13), one has similarly that

$$\overleftarrow{F}_C^f(y_0) \leq \underline{\lim}_{n \rightarrow \infty} \overleftarrow{F}_C^f(y_n). \quad (42)$$

It then follows from (41) and (42) that $\lim_{n \rightarrow \infty} \overleftarrow{F}_C^f(y_n) = \overleftarrow{F}_C^f(y_0)$, and so $\overleftarrow{F}_C^f(\cdot)$ is continuous at y_0 .

Now suppose that C is additionally $\overleftarrow{D}_f$ -Klee and let $y_0 \in U$. To show that $\overleftarrow{Q}_C^f(\cdot)$ is continuous at y_0 , we assume that $\{y_n\} \subseteq U$ with $\lim_{n \rightarrow \infty} y_n = y_0$. Then

$$D_f(\overleftarrow{Q}_C^f(y_n), y_n) = \overleftarrow{F}_C^f(y_n). \quad (43)$$

Noting that $\{\overleftarrow{Q}_C^f(y_n)\} \subseteq C$ and C is compact, there exists a subsequence $\{\overleftarrow{Q}_C^f(y_{n_k})\}$ of $\{\overleftarrow{Q}_C^f(y_n)\}$ such that $\lim_{k \rightarrow \infty} \overleftarrow{Q}_C^f(y_{n_k}) = x_0$ for some $x_0 \in C$. Thus, from (43) and using the continuity of $\overleftarrow{F}_C^f(\cdot)$ at y_0 , we obtain that $D_f(x_0, y_0) = \overleftarrow{F}_C^f(y_0)$. This means $x_0 = \overleftarrow{Q}_C^f(y_0)$ because C is $\overleftarrow{D}_f$ -Klee. Consequently, $\lim_{n \rightarrow \infty} \overleftarrow{Q}_C^f(y_n) = \overleftarrow{Q}_C^f(y_0)$, that is, $\overleftarrow{Q}_C^f(\cdot)$ is continuous at y_0 . The proof is complete. ■

Some sufficient and necessary conditions for a compact $\overleftarrow{D}_f$ -Klee (resp., $\overrightarrow{D}_f$ -Klee) set to be a singleton are the following Theorems 4.6 (resp., Theorem 4.7) below, which extends [3, Theorem 4.4 and Corollary 4.5] and [4, Theorem 4.2(i) $\iff$ (ii)] (resp., [3, Theorem 7.2]) to the setting of this paper.

Theorem 4.6: *Suppose that f is Legendre and that C is compact with $\overline{\text{co } C} \subseteq U$. Then C is a $\overleftarrow{D}_f$ -Klee set if and only if C is a singleton.*

Proof: Clearly, it suffices to show the necessity. To do this, suppose that f is Legendre and that C is compact $\overleftarrow{D}_f$ -Klee with $\overline{\text{co } C} \subseteq U$. It then follows from Proposition 4.4 that $\overline{\text{co } C}$ is a compact $\overleftarrow{D}_f$ -Klee set. Hence, the mapping

$$\overleftarrow{Q}_{\overline{\text{co } C}}^f : \overline{\text{co } C} \rightarrow \overline{\text{co } C}$$

is continuous because $\overleftarrow{Q}_{\overline{\text{co } C}}^f$ is continuous on U thanks to Lemma 4.5. By the Schauder fixed theorem, there exists $x \in \overline{\text{co } C}$ such that $\overleftarrow{Q}_{\overline{\text{co } C}}^f(x) = x$. This implies that

$$\overleftarrow{F}_{\overline{\text{co } C}}^f(x) = D_f\left(\overleftarrow{Q}_{\overline{\text{co } C}}^f(x), x\right) = D_f(x, x) = 0.$$

We assert that $\overline{\text{co } C}$ is a singleton. Otherwise, there is $x' \in \overline{\text{co } C}$, $x' \neq x$. Thus,

$$0 \leq D_f\left(\frac{1}{2}x + \frac{1}{2}x', x\right) < \frac{1}{2}D_f(x, x) + \frac{1}{2}D_f(x', x) \leq \frac{1}{2}\overleftarrow{F}_{\overline{\text{co } C}}^f(x) = 0$$

because $D_f(\cdot, x)$ is strictly convex on U , a contraction. Thus the assertion is shown and so C is a singleton. ■

Theorem 4.7: *Suppose that X is reflexive, and that f is Legendre and Fréchet differential on U . If C is compact with $\overline{\text{co } C} \subseteq U$, then C is a $\overrightarrow{D}_f$ -Klee set if and only if C is a singleton.*

Proof: Suppose that the conditions stated in Theorem 4.7 hold. We only verify the necessary part.

Since X is reflexive and since f is Legendre, it follows from [10, Corollary 5.5] that f^* is Legendre. Hence, f^* is essentially smooth, and so $\nabla f(U) = U^*$ by (11). Therefore, (36) holds. To proceed, let $x^* \in U^*$. Then there is $x \in U$ such that $\nabla f(x) = x^*$. Since C is $\overrightarrow{D}_f$ -Klee, $\overrightarrow{Q}_C^f(x)$ is a singleton, so is $J_X^{-1} \circ \nabla f^* \circ \overleftarrow{Q}_{\nabla f(C)}^{f^*} \circ \nabla f(x)$ by (36). This and Lemma 3.1 imply that $\overleftarrow{Q}_{\nabla f(C)}^{f^*}(x^*) = \overleftarrow{Q}_{\nabla f(C)}^{f^*} \circ \nabla f(x)$ is a singleton, and hence $\nabla f(C)$ is a $\overleftarrow{D}_{f^*}$ -Klee set. Since ∇f is

continuous on U by Proposition 2.2, $\nabla f(C)$ is a compact subset of U^* (noting that C is compact). We assert that

$$\overline{\text{co}(\nabla f(C))} \subseteq U^*. \quad (44)$$

Granting this and Theorem 4.6, $\nabla f(C)$ is a singleton, so is C , and the proof of Theorem 4.7 ends.

Below we show that assertion (44) holds. Since $\nabla f(C) \subseteq U^*$ is compact, there exist finite many points $c_1, \dots, c_n \in C$, and numbers $\delta_1, \dots, \delta_n \in (0, +\infty)$ such that

$$\text{each } \mathbf{B}(\nabla f(c_i), \delta_i) \subseteq U^* \quad \text{and} \quad \bigcup_{i=1}^n \mathbf{B}(\nabla f(c_i), \delta_i) \supseteq \nabla f(C). \quad (45)$$

Noting that X^* is reflexive, each $\mathbf{B}(\nabla f(c_i), \delta_i) \subseteq X^*$ is weakly compact (and convex). Thus, $\text{co}(\bigcup_{i=1}^n \mathbf{B}(\nabla f(c_i), \delta_i))$ is weakly compact (and convex) by [20, Exercise 2.21, p.112], and so weakly closed. By [20, Corollary 1, p.66], $\text{co}(\bigcup_{i=1}^n \mathbf{B}(\nabla f(c_i), \delta_i))$ is closed. This together with (45) implies that

$$\overline{\text{co}(\nabla f(C))} \subseteq \overline{\text{co}(\bigcup_{i=1}^n \mathbf{B}(\nabla f(c_i), \delta_i))} = \text{co}(\bigcup_{i=1}^n \mathbf{B}(\nabla f(c_i), \delta_i)) \subseteq U^*,$$

which completes the proof of (44). ■

The following corollary is a direct consequence of Theorems 4.6 and 4.7.

Corollary 4.8: *Suppose that X is reflexive, and that f is Legendre and Fréchet differential on U . If C is compact with $\overline{\text{co } C} \subseteq U$. Then the following equivalences hold:*

$$C \text{ is a } \overleftarrow{D}_f\text{-Klee set} \iff C \text{ is a } \overrightarrow{D}_f\text{-Klee set} \iff C \text{ is a singleton.}$$

5. Applications to the Euclidean space $\mathbb{R}^n$

In this section, we always assume that

$$X = \mathbb{R}^n, f \text{ is Legendre and } C \text{ is a non-empty closed subset of } \mathbb{R}^n.$$

It is easy to see that C is bounded compact. In order to apply proceeding results to the Euclidean space $\mathbb{R}^n$, we recall that f is 1-coercive (cf.[10]) if

$$\lim_{\|y\| \rightarrow +\infty} \frac{f(y)}{\|y\|} = \infty.$$

From [21, Proposition 2.16], one sees that in $\mathbb{R}^n$, f is 1-coercive if and only if $\text{dom } f^* = \mathbb{R}^n$.

The following Corollaries 5.1 and 5.2 were given in [1, Theorems 4.7 and 7.3] with different proofs.

Corollary 5.1: *Suppose that f is 1-coercive and that C is a $\overleftarrow{D}_f$ -Chebyshev set. Then C is convex.*

Proof: Since f is Legendre, f is essentially smooth. Thus, $\text{dom}(\partial f) = U$ is convex by (10), and so f is strictly convex on U because f is essentially strictly convex. This and [17, 3.1] imply that f is totally convex at any point in U . Therefore, f is locally uniformly totally convex at any point in U by [2, Proposition 2.1] (noting that f is Fréchet differential on U by [22, Theorem 25.1, p.242]). Noting that $\nabla f(U) = U^*$ by (11) and that $\text{dom} f^* = \mathbb{R}^n$, one has that $\nabla f(U) = U^* = \mathbb{R}^n$. Thus, C is convex by Theorem 3.9, which ends the proof of Corollary 5.1. ■

Corollary 5.2: *Suppose that $\text{dom} f = \mathbb{R}^n$ and that $C \subseteq U$ is a $\overrightarrow{D}_f$ -Chebyshev set. Then $\nabla f(C)$ is convex.*

Proof: Similar to the proof of Corollary 5.1, one checks that f is totally convex at any point in $U = \mathbb{R}^n$, and that f^* is locally uniformly totally convex at any point in $\nabla f(X) = U^*$ (see (11)). Thus, $\nabla f(C)$ is convex by Corollary 3.14. ■

Noting that in $\mathbb{R}^n$, if $C \subseteq U$ is a compact set, then $\overline{\text{co } C} = \text{co } C \subseteq U$. Thus, applying above Theorems 4.6 and 4.7 to the Euclidean space $\mathbb{R}^n$, we immediately have the following Corollaries 5.3 and 5.4, which were given respectively in [23, Theorems 1.4.3 and 1.4.2].

Corollary 5.3: *Suppose that C is a compact $\overleftarrow{D}_f$ -Klee set. Then C is a singleton.*

Corollary 5.4: *Suppose that C is a compact $\overrightarrow{D}_f$ -Klee set. Then C is a singleton.*

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